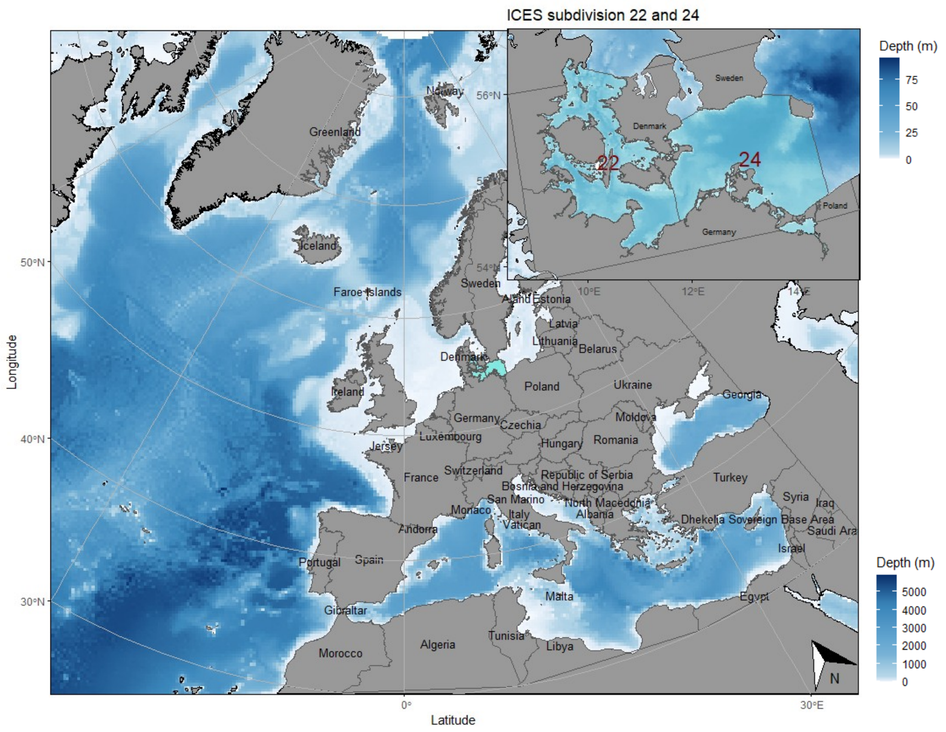
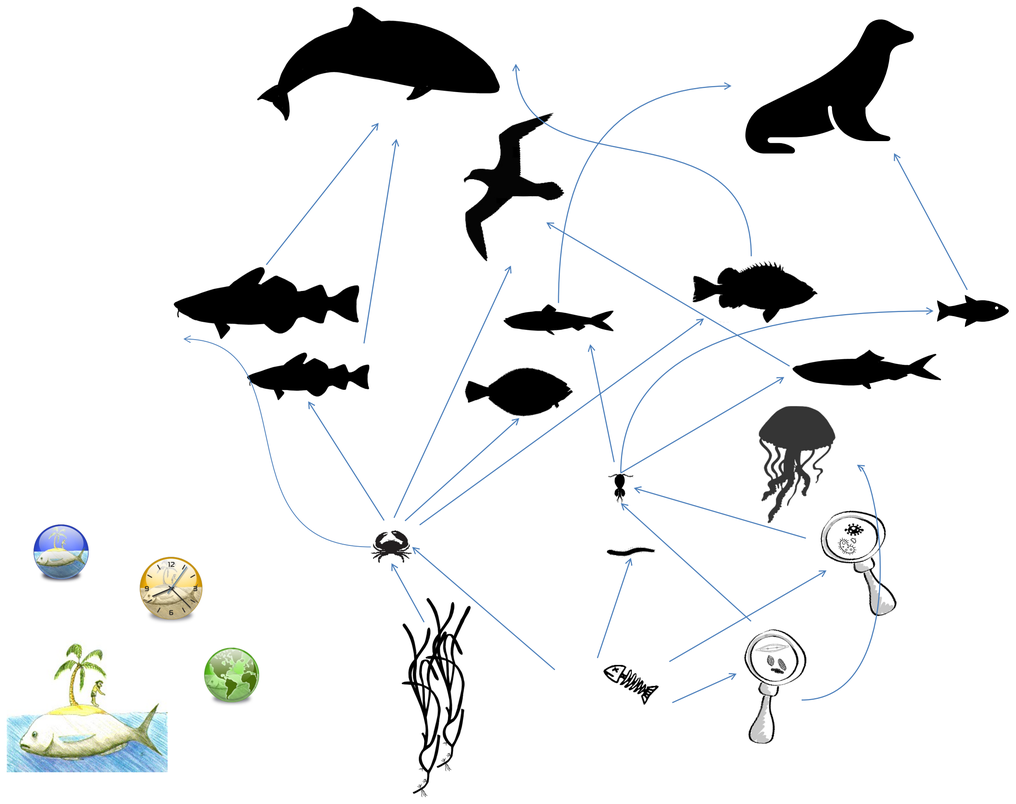
A climate-driven ecosystem model to predict future behavior of trophic groups in the Western Baltic Sea under multiple stressors

Nascimento M.C.[1] and Scotti, M[1]

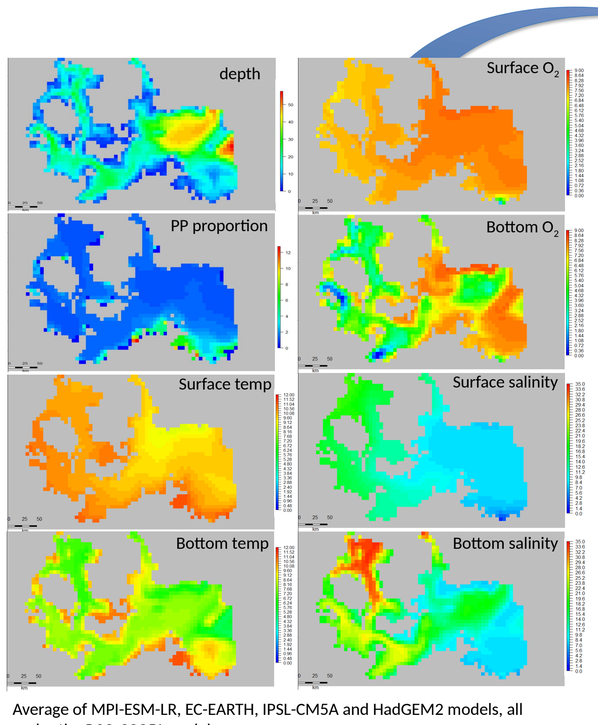
[1] Marine Evolutionary Ecology – RD Marine Ecology, Helmholtz Centre for Ocean Kiel - GEOMAR, Kiel, Germany

We built an Ecopath with Ecosim (EwE) model with 18 functional groups (FG) representing the Wester Baltic Sea (WBS) ecosystem, ICES areas 22 and 24 at the year of 1994 (Scotti et al., 2022).

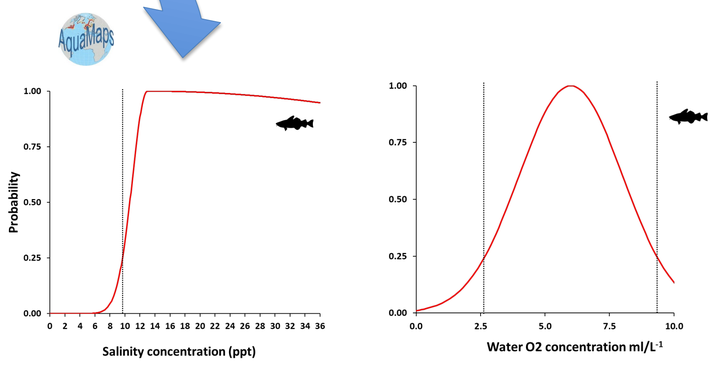




Then we created a spatial model using the Ecospace plugin of the EwE software, including some environmental and habitat layers, as well as environmental limitations (mostly based on AquaMaps) for some functional groups.



Average of MPI-ESM-LR, EC-EARTH, IPSL-CM5A and HadGEM2 models, all under the RCO-SCOBI models



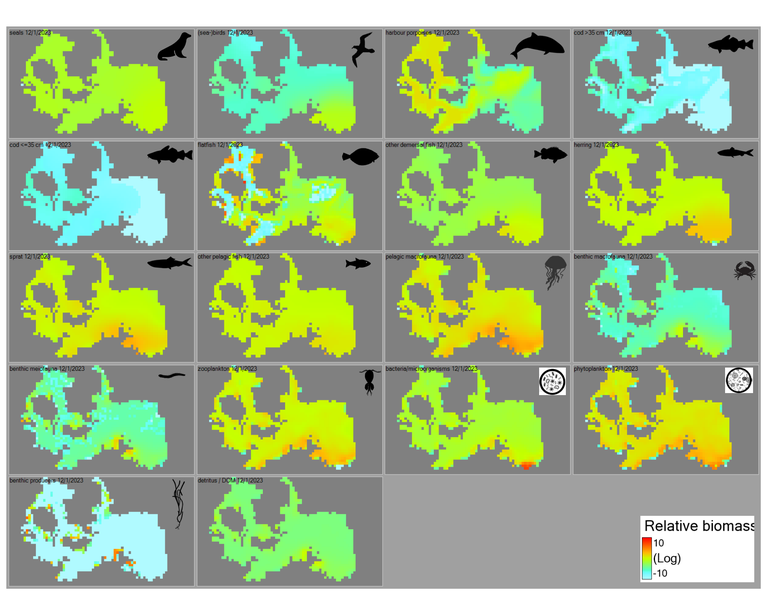
	Temprtaure		O2 concentration		Salinity	
	Bottom	Surface	Bottom	Surface	Bottom	Surface
Harbour porpoise						
Cod adult						
cod juvenile						
Herring						
Sprat						
Flatfishes						

Substrate type



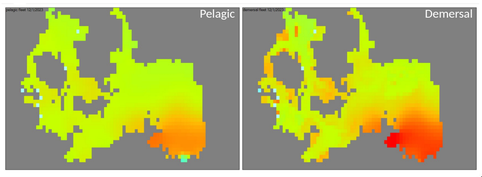
	Hard bot	Bedrock	Sand	Hard clay	Mud	Zostera
Flatfishes						
Benthic macrofauna	0.9	0.8	1	1	1	
Benthic meiofauna	0.5	0.1	1	0.7	1	
Benthic producers	0.2	0.2	1	0.8	1	1

We are working on update the environmental envelope for the FGs for regional data, howeer, our preliminary results, compared to data from monitoring of species distribution, and ICES data on fishing effort shows similarities to the observed distribution and fisheries in the area.

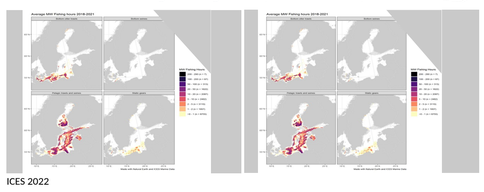


Preliminary distributions – Fishing effort

Ecospace predicted fishing effort



Average fishing effort (vessels > 12 m)



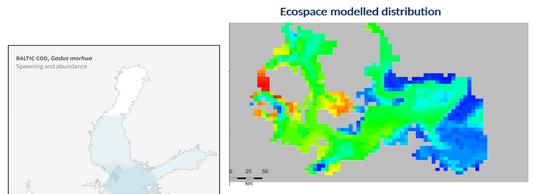
ICES 2022

Preliminary distributions

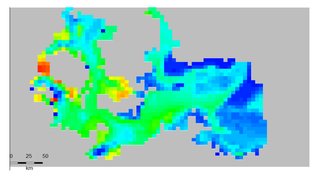
Cod > 35 cm

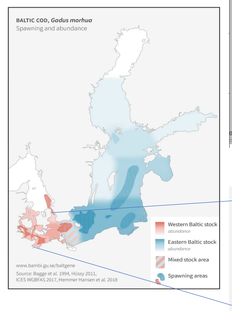


Ecospace modelled distribution

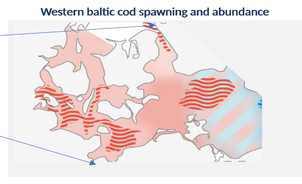


Ecospace predicted catches

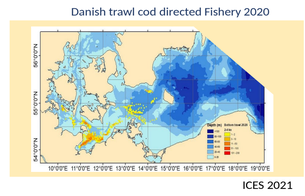




Western baltic cod spawning and abundance



Danish trawl cod directed Fishery 2020



ICES 2021

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